University of the Philippines
VISAYAS

PGC
PHILIPPINE GENOME CENTER
VISAYAS

PGC Visayas Laboratories, Room 205, 2nd Floor, Regional Research Center Building,
Road 8, New Academic Complex, University of the Philippines Visayas, 5023 Miagao, Iloilo

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PGC VISAYAS PRIMER LIST

ORGANISM	PRIMER NAME	TARGET REGION	PRIMER SEQUENCE
ARTIODACTYLA *Bovids, cervids*	Artiodactyla	cytochrome oxidase	*Zlf04_Forward:* TCTCAACTAAYCAYAAAGAYATYGG *Zlr04_Reverse:* TAAACTTCRGGGTGACCAAARAATCA
INSECTA *Lepidoptera, Hymenoptera*	Insects	cytochrome oxidase	*LepF1_Forward:* ATTCAACCAATCATAAAG ATATTGG *LepR1_Reverse:* TAAACTTCTGGATGTCCAAAAAATCA
FISH *Teleosts, Chondrichthyans*	Fish1	cytochrome oxidase	*FishF1_Ward_Forward:* TCAACCAACCACAAAGACATTGGCAC *FishR1_Ward_Reverse:* TAGACTTCTGGGTGGCCAAAGAATCA
	Fish2	cytochrome oxidase	*FishF2_Ward_Forward:* TCGACTAATCATAAAGATATCGGCAC *FishR2_Ward_Reverse:* ACTTCAGGGTGACCGAAGAATCAGAA
AFRICAN SWINE FEVER VIRUS	ASF	Genotype II Strain	*VP72_Forward:* CTGCTCATGGTATCAATCTTATCGA *VP72_Reverse:* GATACCACAAGATCRGCCGT *VP72_Probe:* 56-FAM/CCACGGGAGGAATACCAACCCAGT G/36-TAMSp
ALARIA SPECIES	GAZ2	cytochrome oxidase	*Gaz2_Forward:* CCAACCAYAAAGATATWGGTAC *Gaz2_Reverse:* GGATGACCAAARAACCAAAA
AMPHIBIANS	Amphibians	cytochrome oxidase	*Chmf4_Forward:* TYTCWACWAAYCAYAAAGAYATCGG *Chmr4_Reverse:* ACYTCRGGRTGRCCRAARAATCA
AVIAN	Avian	cytochrome oxidase	*BirdF1_Forward:* TTCTCCAACCACAAAGACATTGGCAC *BirdR1_Reverse:* ACGTGGGAGATAATTCCAAATCCTG

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BACTERIA	16SV1V9	16S	*16S_Forward:* AGAGTTTGATCMTGGCTCAG *16S_Reverse:* TACGGYTACCTTGTTACGACTT
	16S VC	16S	*16SVC_Forward:* ACCTTACCTACTCTTGACATCCA *16SVC_Reverse:* CCCAACATTTCACACACGAG
	Hap	metalloprotease enzyme	*hapA_Forward:* CCAACTTCTTGACCGATCAC *hapR_Reverse:* GCTGGCTTTCAATGTCAGGG
	Qrr	Quorum Regulatory RNAs	*qrr4_Forward:* TGACCCTTCTAAGCCGAGGG *qrr4_Reverse:* GAACAATGGTGTTCACTAACAACG
	16SV3V4	16S	*16SV3_Forward:* TCGTCGGCAGCGTCAGATGTGTATAAGAGA CAGCCTACGGGNGGCWGCAG *16SV3_Reverse:* GTCTCGTGGGCTCGGAGATGTGTATAAGAG ACAGGACTACHVGGGTATCTAATCC
	16SV4	16S	*16SV4_Forward:* TCGTCGGCAGCGTCAGATG TGTATAAGAGACAGGTGCCAGCMGCCGCG GTAA *16SV4_Reverse:* GTCTCGTGGG CTCGGAGATGTGTATAAGAGACAGGGACTA CHVGGGTWTCTAAT
BROWN SEAWEED *(Scytosiphon sp.)*	CAF4	cytochrome oxidase	*Caf4A_Forward:* ATGTTTACTTGGTGRA GRGA *Car4A_Reverse:* CCCCACCARTAWATNGTNAG
	RBCL	rbcl	*rbcL_645_Forward:* ATGCGTTGGAAAGAAAGATTCT *RbcL_R1464_Reverse:* AACATTAGCTGTTGGAGTTTCYAC
RED ALGAE	GAZ	cytochrome oxidase	*GAZF1_Forward:* TCAACAAATCATAAAGATATTGG *GAZR1_Reverse:* ACTTCTGGATGTCCAAAAAAYCA
	C622F	cytochrome oxidase	*C622F_Forward:*

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			CCTGTNTTAGCAGGWGCTATTACAATGC
	F577	rbcl	*F577_Forward:* GTATATGAAGGTCTAAAAGGTGG
	PFG4	ITS	*P5_Forward:* GCATCGATGAAGAACGCAG *G4_Reverse:* CTTTTCCTCCGCTTATTGATATG
Red Algae (*Erythrolobus australicus)*	Plastid	Plastid	*psaA971_Forward:* ACTACWTCATGGCAYGCWCAACT *psaA1110_Reverse:* CCWATCCACATRTGATGTGT
	PSBA	Plastid	*psba_Forward:* ATGACTGCTACTTTAGAAAGACG *psba_Reverse:* GCTAAATCTARWGGGAAGTTGTG
SEAWEED	kcox	cytochrome oxidase	*Kcox3_F3_Forward:* AAGAACCACCTCATCAATA *Kcox2_F71_Forward:* TTCAAGATCCTGCAACTCC *Kcox2_R671_Reverse:* ATTTCACTGCATTGGCCAT *Kcox3_R761_Reverse:* CAYCCTTTCCATTTRGTAGA
	COX1	cytochrome oxidase	*C622F_Forward:* CCTGTNTTAGCAGGWGCTATTACAATGC *Cox1549_Reverse:* AGGCATTTCTTCAAANGTATGATA
	COX2	cytochrome oxidase	*Cox_Forward:* TCAACAAATCATAAAGATATTGGWACT *C880_Reverse:* ACAGTATACATATGATGNGCTCAAAC
	COX3	cytochrome oxidase	*Cox2-3_spacer_Forward:* GTACCWTCTTTDRGRRKDAAATGTGATGC *Cox2-3_spacer_Reverse:* GGATCTACWAGATGRAAWGGATGTC

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CETACEAN	Cetacean1	cytochrome b	*L15601_Forward:* TACGCAATCCTACGATCAATTCC *H15748_Reverse:* GGTTGTCCTCCAATTCATGTTAG
	Cetacean2	cytochrome oxidase	L5357_Forward: CAGTCTAATGCCTACTCGGCC *H6090_Reverse:* GAAGTBTATATYYTAATTCTACC
CRUSTACEAN	Crustacean	cytochrome oxidase	*CrustDF1_Forward:* GGTCWACAAAYCATAAAGAYATTGG *CrustDR1_Reverse:* TAAACYTCAGGRTGACCRAARAAYCA
DIWAL	Diwal	cytochrome oxidase	*Forward:* GGTAGCAGACTCGGGTTGTGG *Reverse:* GCAGAAAACAACAACAAAAGCATAGAA
DUGONG	Dugong	d-loop	*Forward:* CATATTACAACGGTCTTGTAAACC *Reverse:* GTCATAAGTCCATCGAGATGTC
FISH	Ecoprimer	12S	*Forward:* TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG [locus-specific sequence] ACTGGGATTAGATACCCC *Reverse:* GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG [locus-specific sequence]TAGAACAGGCTCCTCTAG
	FishU	16S	*Forward:* TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG-[locus-specific sequence] ACAYACCGC CCGTCACYCTC *Reverse:* GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG-[locus-specific sequence] CAT GAT GCA AAA GGT ACR RG

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FISH	Mifish	12S	*Mifish_Forward:* TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG GTCGGTAAAACTCGTGCCAGC *Mifish_Reverse:* GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG CATAGTGGGGTATCTAATCCCAGTTTG
	Fish3	cytochrome oxidase	*VF2_t1_Forward:* TGTAAAACGACGGCCAGTCAACCAACCACAAAGACATTGGCAC *FishF2_t1_Reverse:* TGTAAAACGACGGCCAGTCGACTAATCATAAAGATATCGGCAC *FishR2_t1_Forward:* CAGGAAACAGCTATGACACTTCAGGGTGACCGAAGAATCAGAA *FR1d_t1_Reverse:* CAGGAAACAGCTATGACACCTCAGGGTGTCCGAARAAYCARAA
	Fish	cytochrome oxidase	*FishF1_Liu:* TCTCAACCAACCATAAAGACATTGG *FishR1_Liu:* TATACTTCTGGGTGCCCAAAGAATCA *Fish2_Forward:* CATCCTACCTGTGGCAATCAC *Fish2_Reverse:* TGGGCTCAGACAATAAATCCT
FUNGI	RPB2	rpb2	*fRPB2–7cR_Reverse:* CCCATRGCTTGYTTRCCCAT *fRPB2–5F_Forward:* GAYGAYMGWGATCAYTTYGG
	TEF	trans elongation factor	*983_Forward:* GCYCCYGGHCAYCGTGAYTTYAT *2218_Reverse:* ATGACACCRACRGCRACRGTYTG
	LSU	large sub unit	*LROR_Forward:* GTACCCGCTGAACTTAAGC *LR5_Reverse:* ATCCTGAGGGAAAC TC

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FUNGI	Actin	Actin	*ACT512_Forward:* ATGTGCAAGGCCGGTTTCGC *ACT783_Reverse:* TACGAGTCCTTCTGGCCCAT
	Tubulin	beta tubulin	*T1_Forward:* AACATGCGTGAGATTGTAAGT *BT2B_Reverse:* ACCCTCAGTGTAGTGACCCTTGGC
	TEF	trans elongation factor	*983_Forward:* GCYCCYGGHCAYCGTGAYTTYAT *2218_Reverse:* ATGACACCRACRGCRACRGTYTG
	LSU	large sub unit	*LROR_Forward:* GTACCCGCTGAACTTAAGC *LR5_Reverse:* ATC CTG AGG GAA ACT TC
	ITS Fungi	ITS	ITS1_Forward: CTTGGTCATTTAGAGGAAGTAA ITS5_Forward: GGAAGTAAAAGTCGTAACAAGG ITS4_Reverse: TCCTCCGCTTATTGATATGC
GREEN MUSSEL	Beta Actin	Beta Actin	*beta_Forward:* CGACGACAGCAGAAAGAG *beta_Reverse:* CGTCAGGCAGTTCGTAAC
	HeatShock	Heatshock protein	*HS_Forward:* CCAACTGGTGGTGCTGG *HS_Reverse:* CAAATTAACATCTGCAACACTGATA
	Dloop	d-loop	*dloopf1_Forward:* GACTCGGCAAGAAGAAGTTG *dloopf2_Forward:* GAGGGATAGGCTAGGTTGTATATTG *dloopf3_Forward:* GGAGCTAAGATGTTAAGAATATAAGGAT
	Nad	NADH	*nad4_Reverse:* CCCATTCTAATTACCTCATCATTGA *nad4lau_Reverse:* TGCCACATAAACTACCCTCATC

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INVERTEBRATES	Lernaea1	cytochrome oxidase	Forward: TAGTTGGAATTTGGGCTGGC *Reverse:* ATTAGGGGCCTTGTTGGGAAG
	Lernaea2	18S	*Forward:* AAGGTGTGMCCTATCAACT *Reverse:* TTACTTCCTCTAAACGCTC
	Folmer	cytochrome oxidase	*LCO1490_Forward:* GGTCAACAAATCATAAAGATATTGG *HCO2198_Reverse:* TAAACTTCAGGGTGACCAAAAAATCA
MACROBRACHIUM	Macrobrachium1	cytochrome oxidase	*Forward:* GCTTTTYTCAACAAACCATAAAGACATTGG *Reverse:* TGTCACTCGATAGARGTTGGTATRAA
MARINE INVERTEBRATES	Myt18S	cytochrome oxidase	*Forward:* CAACCTGGTTGATCCTGCCAGT *Reverse:* CTGATCCTTCTGCAGGTTCACCTAC
MICROALGAE	TUF	trans elongation factor	*tufA-F (Fama)_Forward:* GGNGCNGCNCAAATGGAYGG *tufA-R (Fama)_Reverse:* CCTTCNCGAATMGCRAAWCGC
	Algae1	18S	*ss5_Forward:* GGTGATCCTGCCAGTAGTCATATGCTTG *ss3_Reverse:* GATCCTTCCGCAGGTTCACCTACGGAAACC
	Algae2	ITS	*its_Forward:* AGGAGAAGTCGTAACAAGGT *its_Reverse:* TCCTCCGCTTATTGATATGC

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PLANT Angiosperms, gymnosperms, ferns	PSBA1-trnH	Chloroplast gene	*psbA_Forward:* GTWATGCAYGAACGTAATGCTC *trnH_Reverse:* CGCGCATGGTGGATTCACAATCC
	RBCL1	Plastid	*rbcl_1F_Forward:* ATGTCACCACAAACAGAAAC *rbcl_724R_Reverse:* TCGCATGTACCTGCAGTAGC
	MatK	Plastid	*matK_472f_Forward:* CCCRTYCATCTGGAAATCTTGGTT *matK_1248R_Reverse:* GCTRTRATAATGAGAAAGATTTCTGC
	ITS2	ITS2	*ITS2_S2F_Forward:* ATGCGATACTTGGTGTGAAT *ITS2_S3R_Reverse:* GACGCTTCTCCAGACTACAAT
	PS1	Photosystem 1 chlorophyll a apoprotein A1	*psaA130_Forward:* AACWACWACTTGGATTTGGAA *psaA1760_Reverse:* CCTCTWCCWGGWCCATCRCAWGG
REPTILE	RepCOI	Cytochrome oxidase I	*Forward:* TNTTMTCAACNAACCACAAA A *Reverse:* ACTTCTGGRTGKCCAAARAATCA
SHIPWORM	18S-Euk shw	18S	*Forward:* WAYCTGGTTGATCCTGCCAGT *Reverse:* TGATCCTTCYGCAGGTTCACCTAC
	28S-NL1600 shw	28S	*Forward:* ACCCGCTGAAYTTAAGCATAT *Reverse:* AGCGCCATCCATTTTCAGG
THRAUSTOCHYTRIDS	Thrausto1	ITS	*F-Thrausto_Forward:* GGGAGCCTGAGAGACGGC *R-Thrausto_Reverse:* GGCCATGCACCACCACCC
	Thrausto2	18S	*18S001_Forward:* AACCTGGTTGATCCTGCCAGTA *18S13_Reverse:* CCTTGTTACGACTTCACCTTCCTCT

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WARTY PIG	Uniminibar	cytochrome oxidase	*Forward:* TCCACTAATCACAARGATATTGGTAC *Reverse:* GAAAATCATAATGAAGGCATGAGC
YEAST	ITS2F with OH	ITS	*ITS2F with OH:* TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGGCATCGATGAAGAACGC *ITS2R with OH:* GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGCTCCGCTTATTGATATGC